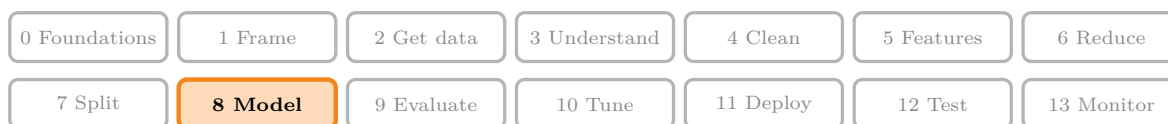


The Transformer Encoder

Contents

1. Overview	2
2. Prerequisites	3
3. From a black box to the real block	4
4. The input: from words to vectors	5
5. Inside one encoder block	5
5.1 Multi-head attention	6
5.2 Add and norm	6
5.3 The feed-forward network	7
5.4 Add and norm again, then the next block	9
6. Six blocks: shapes and parameters	11
7. Why each part is there	12
7.1 Why residual connections	12
7.2 Why the feed-forward network	14
7.3 Why 6 blocks	15
8. Summary	17
9. Sources	17
10. Key terms	18

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Sequence-to-sequence (encoder-decoder) ([Note DL-058](#)); Multi-head attention ([Note DL-078](#)); Positional encoding ([Note DL-079](#)); Layer normalisation ([Note DL-080](#)); Transformer decoder ([Note DL-084](#)); Learning-rate warm-up schedule ([Note DL-086](#)).
- **Leads to:** Transformer decoder ([Note DL-084](#)); The transformer end to end (capstone) ([Note DL-086](#)); Decoder-only GPT ([Note DL-087](#)).
- **Compare with:** LSTM (long short-term memory) ([Note DL-063](#)).

1. Overview

Key point: The transformer has two parts, an **encoder** (G-682) and a **decoder** (G-564). The encoder turns a sentence into one context-aware vector per word. It first adds a positional encoding to each word's embedding, then passes the vectors through 6 identical **encoder blocks** (G-681). Each block has two **sub-layers** (G-1908): multi-head attention, then a small **feed-forward network** (G-775) that works on each word separately. Each sub-layer is wrapped in “add and norm”: a **residual connection** (G-1681) adds the sub-layer's input to its output, and a layer normalisation follows. Every vector keeps 512 numbers from start to end.

The earlier Notes built the parts one at a time: self-attention, its scaling, multi-head attention, positional encoding and layer normalisation. This Note puts them together into the encoder of “Attention Is All You Need” (Vaswani et al. 2017, §3.1). The decoder is the subject of the [transformer decoder Note](#).

Figure 1 is the whole architecture. The encoder side is drawn in full. The decoder side is greyed out here.

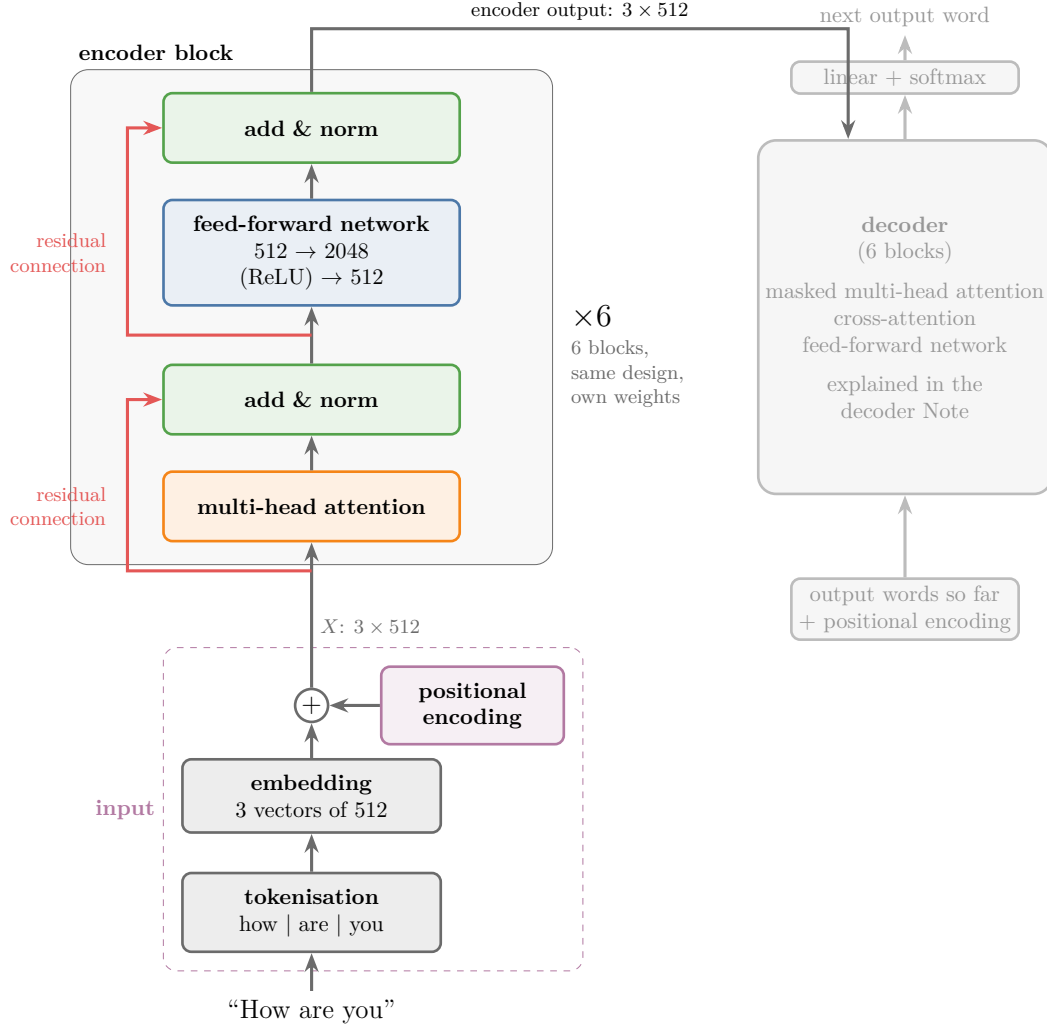


Figure 1: The transformer. Left: the input steps and one encoder block, repeated 6 times; red arrows are the residual connections. Right (grey): the decoder, which receives the encoder's output

We follow one sentence, "How are you", through the encoder. Along the way we:

- check a by-hand computation against Keras (section 5);
- count shapes and parameters (section 6);
- test why each part is there (section 7).

2. Prerequisites

- **Multi-head attention Note:** 8 heads, concatenation, W_O ; one 512-number output per word.
- **Positional encoding Note:** a 512-number vector per position, added to the word's embedding.

- **Layer normalisation Note:** each word’s vector normalised by its own mean and standard deviation, then scaled by γ and shifted by β .
- **Self-attention step by step Note:** contextual embeddings from queries, keys and values.
- **Forward propagation Note:** a dense layer as a matrix product plus a bias, followed by an activation.
- **Vanishing and exploding gradients Note:** why deep networks are hard to train.

3. From a black box to the real block

Key point: The transformer is an encoder and a decoder. The encoder is 6 identical blocks in a row. Each block is attention followed by a feed-forward network, with add and norm after each.

The full diagram looks complicated, so we build it up in four steps (Figure 2).

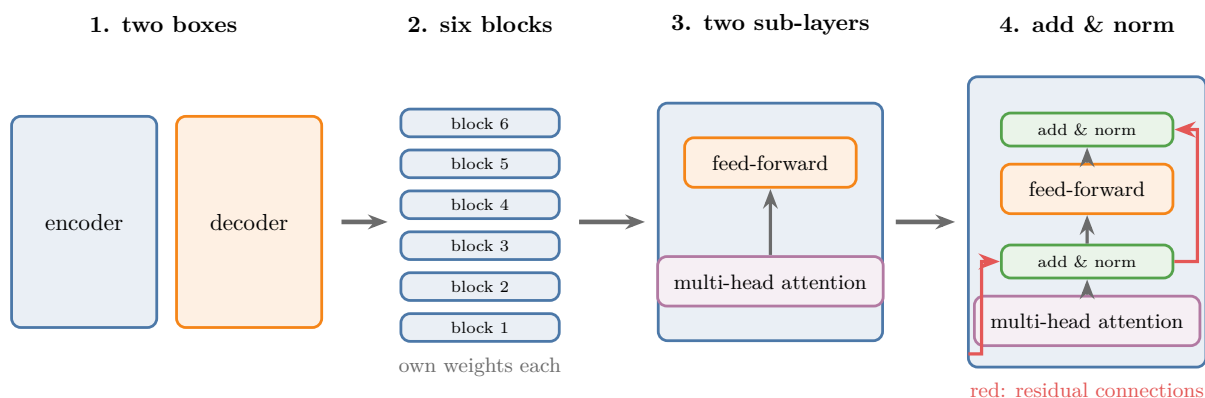


Figure 2: The encoder built up in four steps: two boxes; six blocks in the encoder; two sub-layers in each block; an add-and-norm step after each sub-layer, with the residual connections in red

1. **Two boxes.** A transformer is one box holding two boxes: the encoder and the decoder. The encoder reads the input sentence; the decoder writes the output sentence.
2. **Six blocks each.** Next to both boxes the paper’s diagram writes “ $N \times$ ”. The encoder is a stack of $N = 6$ identical layers, and so is the decoder (Vaswani et al. 2017, §3.1). We call each layer an **encoder block**.
3. **Two sub-layers per block.** Each encoder block has two parts: a multi-head self-attention, then a feed-forward network (Vaswani et al. 2017, §3.1).
4. **Add and norm.** Around each of the two sub-layers sits a residual connection, followed by a layer normalisation: the “add & norm” boxes of Figure 1.

“Identical” means the same design. Every block has its own weights, learned separately (section 6).

The sentence enters at the bottom of the first block. The first block’s output is the input of the second block, and so on. The output of the sixth block goes to the decoder.

4. The input: from words to vectors

Key point: Three steps happen before the first block: split the sentence into tokens, look up a 512-number embedding for each token, and add the positional encoding of its position. The result is the input matrix X , one row per word.

The purple box of Figure 1 does three things to “How are you”:

1. **Tokenisation.** The sentence is split into **tokens** (G-1981), the units the model reads. With word-level tokenisation, each token is a word: “how”, “are”, “you”. The paper itself uses pieces of words, a method called byte-pair encoding (Vaswani et al. 2017, §5.1).
2. **Embedding.** Each token becomes a learned vector of $d_{\text{model}} = 512$ numbers (see the [word embeddings section](#) of the RNN sentiment Note).
3. **Positional encoding.** The vector of each position, also 512 numbers, is added to the word’s embedding, so that the model knows the word order (the [positional encoding Note](#)).

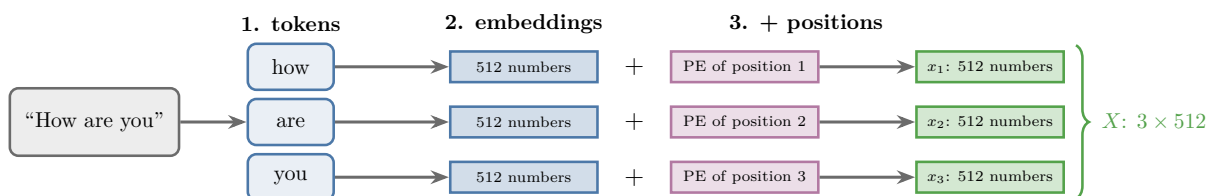


Figure 3: The three input steps for “How are you”: the sentence is split into tokens, each token gets a 512-number embedding, and the positional encoding of its position is added. The three sums are the rows of X

Figure 3 follows the three steps for each word: a token is the same embedding wherever it appears, and only the added positional encoding tells the model where it stands.

The three resulting vectors x_1, x_2, x_3 are stacked as the rows of a 3×512 matrix X . In practice a batch of many sentences goes in together; here the batch holds one sentence.

5. Inside one encoder block

Key point: A block does four things in order: attention lets the words exchange information, an add-and-norm step adds the input back and rescales, a feed-forward network transforms each word on its own, and a second add-and-norm step follows. In symbols: $X \rightarrow \text{multi-head attention} \rightarrow Z$; add X and normalise $\rightarrow Z_{\text{norm}}$; feed-forward network $\rightarrow Y$; add Z_{norm} and normalise $\rightarrow Y_{\text{norm}}$. Every one of these matrices is 3×512 , except the hidden layer of the feed-forward network, which is 3×2048 .

Figure 4 shows the path of the three words through one block, with every shape.

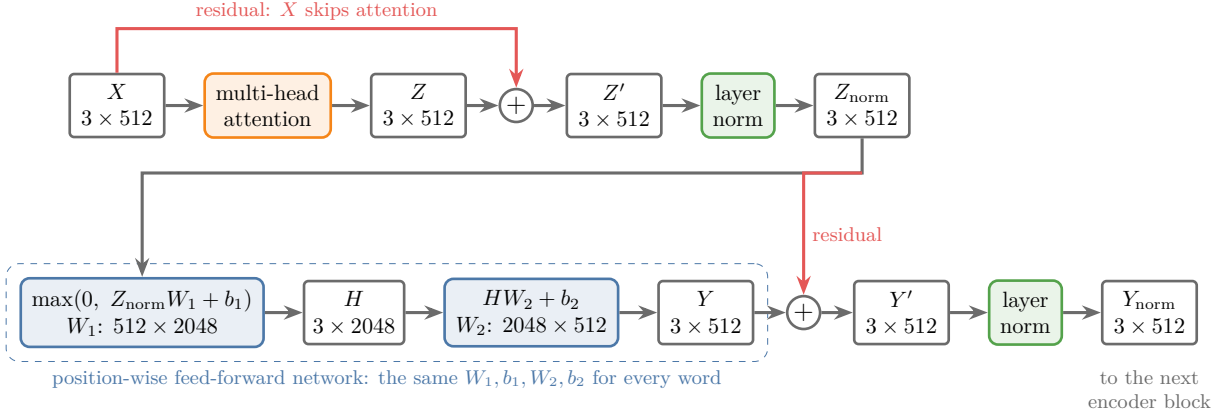


Figure 4: One encoder block for a 3-word sentence. Every matrix is 3×512 except H , the hidden layer of the feed-forward network. The red arrows are the residual connections

To show real numbers, the Notebook builds a tiny block: $d_{\text{model}} = 4$, 2 heads of 2 numbers, and a feed-forward hidden layer of 8. The embeddings of “how”, “are”, “you” are made-up values, and the block’s weights are Keras’ random starting weights. The steps and the formulas are exactly those of the full-size block. We follow the word “how”, whose input row, embedding plus positional encoding, is

$$x_{\text{how}} = [0.5, 1.0, -0.5, 0.2] + [0, 1, 0, 1] = [0.5, 2.0, -0.5, 1.2]$$

5.1 Multi-head attention

Key point: Multi-head attention turns each word’s vector into a contextual vector of the same size, using every word of the sentence.

The first sub-layer is the multi-head attention of the [multi-head attention Note](#), with queries, keys and values all taken from X . The embedding of a word is the same in every sentence; attention mixes in the other words, so “bank” in “river bank” gets a different vector from “bank” in “money bank”. Its output Z has one 512-number row per word, z_1, z_2, z_3 , each aware of the whole sentence.

For “how” in the tiny block:

$$z_{\text{how}} = [1.08, 0.77, 0.39, -1.21]$$

5.2 Add and norm

Key point: Add the sub-layer’s input to its output (the residual connection), then apply layer normalisation to each word: $\text{LayerNorm}(x + \text{Sublayer}(x))$.

The paper writes the output of each sub-layer as $\text{LayerNorm}(x + \text{Sublayer}(x))$ (Vaswani et al. 2017, §3.1). Two things happen.

Add. Besides going through the attention, X also takes a second path that skips the attention (the red arrow). At the “add” box, the two meet: $Z' = X + Z$. The addition works because both are 3×512 ; the paper keeps every sub-layer’s output at $d_{\text{model}} = 512$ “to facilitate these residual connections” (Vaswani et al. 2017, §3.1). A path that skips a sub-layer and is added back is a **residual connection** (also called a skip connection). Section 7.1 explains why it is there.

Norm. Each row of Z' is then normalised on its own: subtract the row’s mean, divide by its standard deviation, multiply by γ and add β (the [layer normalisation Note](#)). The result is Z_{norm} .

1. **In words:** add the word’s input to its attention output, then standardise the 4 (in the full model, 512) numbers of the sum.
2. **Formula:**

$$z'_i = x_i + z_i, \quad z_{\text{norm},i} = \gamma \odot \frac{z'_i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}} + \beta$$

where μ_i and σ_i^2 are the mean and variance of the numbers in z'_i , and $\odot$ multiplies number by number.

3. **Example:** for “how”,

$$z'_{\text{how}} = [0.5 + 1.08, 2.0 + 0.77, -0.5 + 0.39, 1.2 - 1.21] = [1.58, 2.77, -0.11, -0.01]$$

with mean $\mu = 1.06$ and standard deviation $\sigma = 1.20$. With Keras’ starting values $\gamma = 1$, $\beta = 0$:

$$z_{\text{norm,how}} = \left[\frac{1.58-1.06}{1.20}, \frac{2.77-1.06}{1.20}, \frac{-0.11-1.06}{1.20}, \frac{-0.01-1.06}{1.20} \right] = [0.44, 1.43, -0.98, -0.89]$$

Attention as a change. Read the addition the other way round: the word’s vector x stays, and attention adds a change z to it. The [self-attention geometrically Note](#), section 7.3, draws this as an arrow e with a change Δe placed at its tip, and measures how small the change is in a trained model (Sanderson 2024, Ch 6). The feed-forward network of section 5.3 adds a second change in the same way (Figure 6).

Why normalise here. The outputs of attention have no fixed range, and adding X can make them larger still. Training is more stable when the numbers stay in a small range; layer normalisation brings every word’s vector back to mean 0 and standard deviation 1 before the next sub-layer (Ba et al. 2016; the [layer normalisation Note](#)).

5.3 The feed-forward network

Key point: A two-layer network, $512 \rightarrow 2048$ with ReLU $\rightarrow 512$ with no activation, applied to each word’s vector separately with the same weights: $\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$.

The second sub-layer is a small fully connected network (Vaswani et al. 2017, §3.3):

Layer	Nodes	Activation	Weights	Biases
input	512	—	—	—
hidden	2048	ReLU	$W_1: 512 \times 2048$	$b_1: 2048$
output	512	none (linear)	$W_2: 2048 \times 512$	$b_2: 512$

1. **In words:** widen each word’s vector from 512 to 2048 numbers, set the negative ones to 0 (ReLU), then bring it back to 512 numbers.
2. **Formula:** for the whole matrix at once,

$$H = \max(0, Z_{\text{norm}}W_1 + b_1) \quad (3 \times 2048), \quad Y = HW_2 + b_2 \quad (3 \times 512)$$

3. **Example:** in the tiny block ($4 \rightarrow 8 \rightarrow 4$), the row of “how” becomes

$$h_{\text{how}} = [0.20, 0, 0.83, 0.09, 0.21, 1.36, 0, 0.47], \quad y_{\text{how}} = [-0.78, 0.67, -0.14, -0.31]$$

Two of the eight hidden values are 0: ReLU cut them off.

The three rows of Z_{norm} go in together, like a batch of 3 observations (records) for an ordinary network. Each row is still processed on its own. The paper calls the network **position-wise** (G-1527): it is “applied to each position separately and identically” (Vaswani et al. 2017, §3.3). The Notebook checks both halves of that sentence:

- **Separately:** feeding the three rows one at a time gives exactly the same output as feeding the whole matrix. Changing the vector of “you” changes the output of “you” (by up to 7.13) and leaves the outputs of “how” and “are” exactly as they were.
- **Identically:** the same W_1, b_1, W_2, b_2 serve every position.

Words exchange information only in the attention sub-layer: the same change to “you” alters the attention output of every word (by 1.59, 4.12 and 5.75 for “how”, “are”, “you”).

change in each word's output when only "you" changes

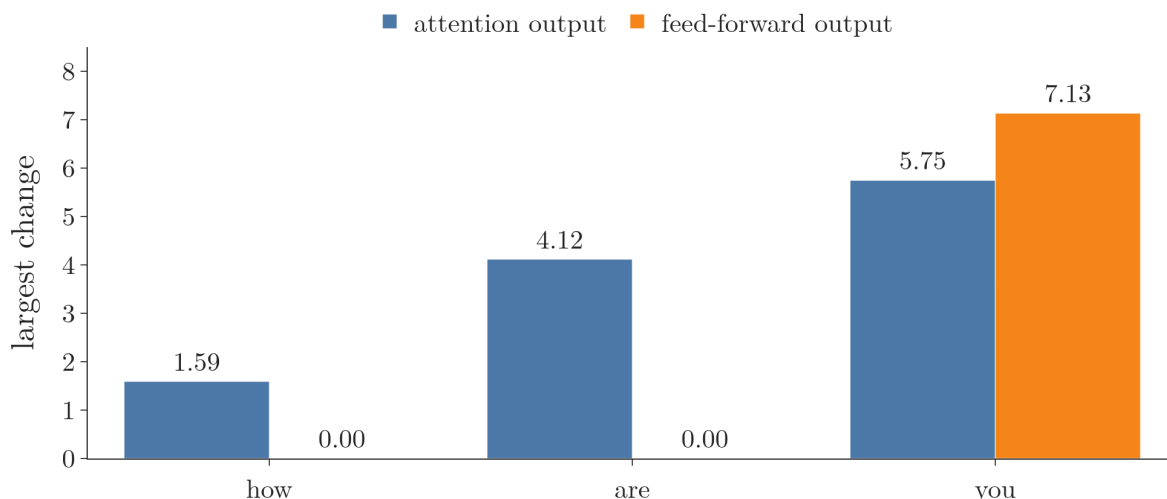


Figure 5: The tiny block of this section, with 5 added to every number of “you”. Blue: how much each word’s attention output changes. Orange: how much each word’s feed-forward output changes

In Figure 5, the orange bars of “how” and “are” are exactly 0: the feed-forward network never sees “you” when it processes them. The blue bars show attention passing the change to every word.

Extra: One way to read W_1 and W_2 (Sanderson 2024, Ch 7; Geva et al. 2021):

- **W_1 asks questions.** Each hidden value is the dot product of the word’s vector with one column of W_1 , plus a bias: how strongly the word has one particular feature.
- **ReLU answers yes or no.** A hidden value that is negative becomes 0, so that question contributes nothing.
- **W_2 writes the answers.** The output $y = hW_2$ is the sum of the rows of W_2 , each scaled by its hidden value: every active hidden value adds its own row, a direction, to the word’s vector.

We write the word as a row vector times the matrix, xW_1 , as the paper does. Sanderson writes the matrix times a column vector, so for him the questions are the *rows* of the first matrix and the written directions are the *columns* of the second. The [Note on how the feed-forward network stores facts](#) builds this reading up on a real model.

5.4 Add and norm again, then the next block

Key point: The feed-forward output gets its own residual connection and layer normalisation. The result, still 3×512 , is the input of the next block.

The second add and norm repeats section 5.2, with Z_{norm} as the input that skips the sub-layer:

$$Y' = Z_{\text{norm}} + Y, \quad Y_{\text{norm}} = \text{LayerNorm}(Y')$$

For “how”: $y'_{\text{how}} = [0.44 - 0.78, 1.43 + 0.67, -0.98 - 0.14, -0.89 - 0.31] = [-0.34, 2.10, -1.12, -1.20]$, and after normalisation $y_{\text{norm,how}} = [-0.15, 1.68, -0.73, -0.79]$.

Y_{norm} is the block’s output. It plays the role of X for the second block, which runs the same steps with its own weights.

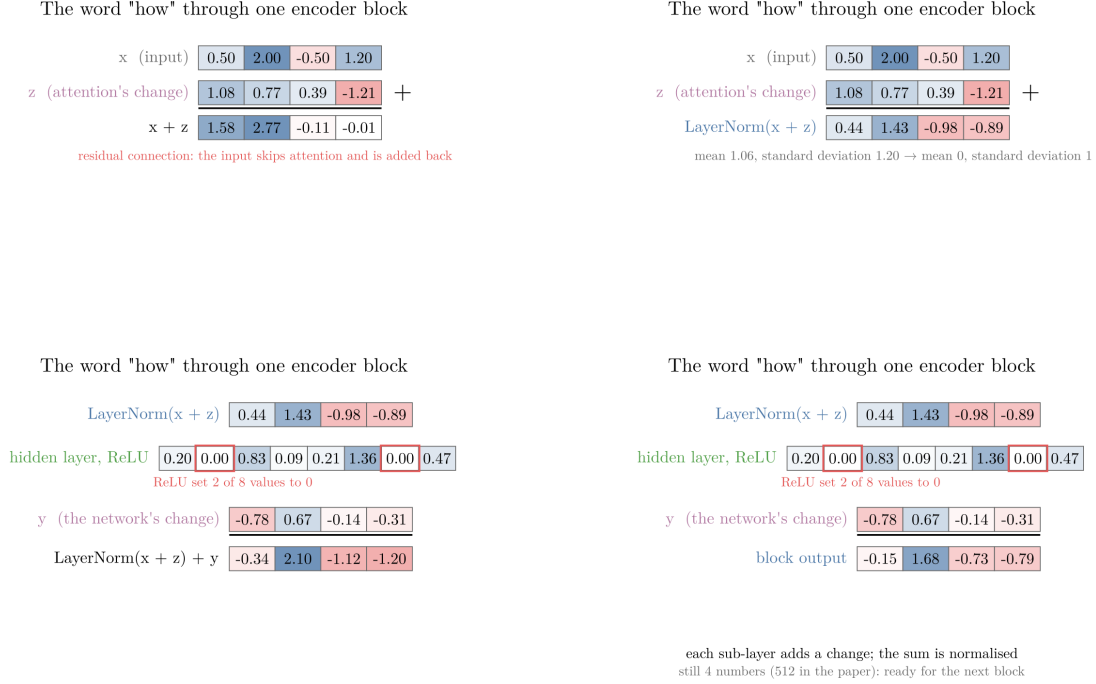


Figure 6: The word “how” through the Notebook’s tiny encoder block. Attention computes a change z that is added to x ; layer normalisation rescales the sum; the feed-forward network computes a second change y that is added; a second normalisation gives the block’s output. Blue cells are positive, red negative

Watch in Figure 6 how each sub-layer only adds a change to the vector that comes in; the normalisation then brings the sum back to mean 0 and standard deviation 1.

The Notebook repeats the whole block by hand in NumPy, from X to Y_{norm} , and compares with the Keras block built from `MultiHeadAttention`, `LayerNormalization` and `Dense` layers. The largest difference is 5×10^{-7} : rounding error.

Python: One encoder block as a Keras layer (post-norm, as in the paper).

```
class EncoderBlock(keras.layers.Layer):
    def __init__(self, d_model, heads, d_ff):
        super().__init__()
        self.mha = keras.layers.MultiHeadAttention(
```

```

        num_heads=heads, key_dim=d_model // heads)
self.norm1 = keras.layers.LayerNormalization()
self.ffn1 = keras.layers.Dense(d_ff, activation="relu")
self.ffn2 = keras.layers.Dense(d_model)
self.norm2 = keras.layers.LayerNormalization()

def call(self, x):
    # attention, then add & norm
    z_norm = self.norm1(x + self.mha(x, x))
    # feed-forward, then add & norm
    y = self.ffn2(self.ffn1(z_norm))
    return self.norm2(z_norm + y)

```

Extra: The paper also applies dropout (the [dropout Note](#)) with a rate of 0.1 to the output of each sub-layer, before it is added to the sub-layer’s input, and to the sum of the embeddings and positional encodings (Vaswani et al. 2017, §5.4, “Residual Dropout”). Dropout only acts during training; the block above leaves it out to keep the computation exact.

6. Six blocks: shapes and parameters

Key point: The shape stays $n \times 512$ through all 6 blocks. One block has 3,152,384 parameters, two-thirds of them in the feed-forward network; the encoder’s 6 blocks have 18,914,304.

Shapes. Every sub-layer maps 3×512 to 3×512 , so the 6 blocks can be chained, and the encoder’s output is again 3×512 : one vector per input word, now informed by the whole sentence. The Notebook passes a 3×512 input through 6 Keras blocks and gets 3×512 out. For a sentence of n words, read n for 3.

Parameters. The counts follow from the layer sizes, with $d = 512$ and $d_{\text{ff}} = 2048$ (the attention formula is derived in the [multi-head attention Note](#), section 6.1):

Part	Formula	Parameters
Multi-head attention	$4d^2 + 4d$	1,050,624
Feed-forward network	$d \cdot d_{\text{ff}} + d_{\text{ff}} + d_{\text{ff}} \cdot d + d$	2,099,712
Two layer normalisations	$2 \times 2d$ (γ and β)	2,048
One encoder block		3,152,384
6 encoder blocks		18,914,304

Keras’ `count_params()` gives exactly these numbers (Notebook). The embedding layer comes on top and depends on the vocabulary size.

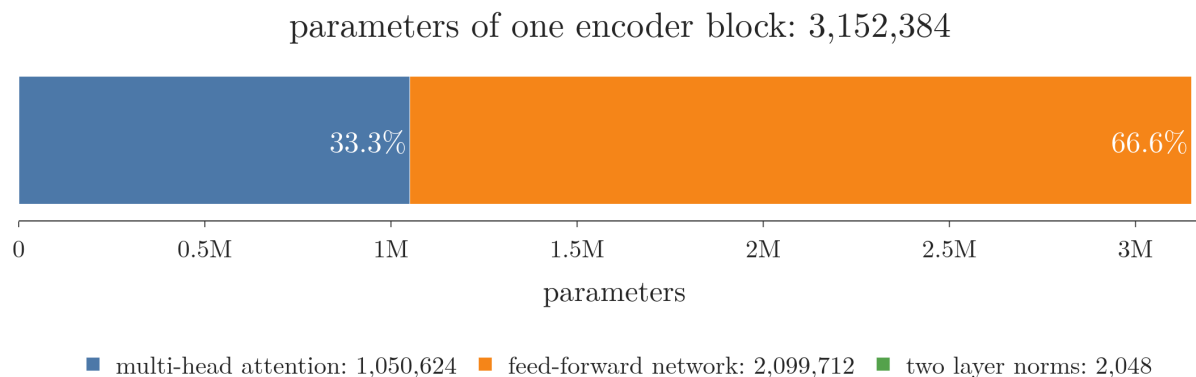


Figure 7: One encoder block’s parameters split by part. The two layer normalisations hold 2,048, too few to show

The feed-forward network holds 2,099,712 of a block’s 3,152,384 parameters (Figure 7): 66.6 percent, two-thirds. Geva et al. (2021) open their study of these layers with the same observation: “feed-forward layers constitute two-thirds of a transformer model’s parameters”.

Same design, own weights. The 6 blocks are copies of one design, not of one set of numbers. Each block has its own attention matrices, its own W_1, b_1, W_2, b_2 and its own γ, β , all updated separately by backpropagation. The paper says this of the feed-forward networks: the transformations “are the same across different positions”, but “they use different parameters from layer to layer” (Vaswani et al. 2017, §3.3). In the Notebook, the first entry of W_1 is 0.038 in block 1 and -0.034 in block 2.

7. Why each part is there

Key point: Residual connections keep a direct path for the input and for the gradient through a deep stack. The feed-forward network adds a non-linear transformation of each word, and holds most of the parameters. Several blocks give the model the depth to represent language; 6 worked best among the sizes the paper tried.

The paper describes the encoder but says little about why each part was chosen. The reasons below come from the papers that introduced the parts, and from experiments.

7.1 Why residual connections

Key point: A residual connection lets a sub-layer’s input pass straight through, so the sub-layer only has to learn a change to its input. In deep networks this makes training easier, and it keeps each word’s own information from being lost. Without residual connections, a single untrained encoder block gave all 30 words of a review the same vector.

The paper uses residual connections without explaining the choice; it cites the network that introduced them, ResNet (He et al. 2016a). Two reasons are given in the literature.

Reason 1: deep stacks train more easily. He et al. (2016a) observed a **degradation problem** (G-576): when they added layers to an already deep plain network, the training error went up, not down, and the cause was not overfitting. A deeper network should never be worse on the training data, since its extra layers could simply copy their input (an identity mapping). Plain layers find copying hard. With a residual connection, a layer computes $F(x) + x$, and copying only requires $F(x) = 0$: “the solvers may simply drive the weights of the multiple nonlinear layers toward zero to approach identity mappings” (He et al. 2016a, §3.1).

1. **In words:** if a sub-layer has nothing useful to add, its weights can go to 0, and the input passes through unchanged apart from the normalisation.
2. **Formula:** with all weights and biases of the sub-layer at 0, $\text{Sublayer}(x) = 0$, so

$$\text{LayerNorm}(x + \text{Sublayer}(x)) = \text{LayerNorm}(x)$$

Without the residual connection the same block would output $\text{LayerNorm}(0)$, which is just β for every word.

3. **Example:** for “how” in section 5.2, $x = [0.5, 2.0, -0.5, 1.2]$ would come out as $\text{LayerNorm}(x) = [-0.33, 1.30, -1.41, 0.43]$ (Notebook), still “how”. Without the residual connection it would be $[0, 0, 0, 0]$ (with $\beta = 0$), the same for every word.

A follow-up paper showed a second effect of the identity path: the gradient of the loss with respect to an early layer contains a term that comes straight from the last layer, without passing through any weights, so it “is unlikely” to vanish (He et al. 2016b, §3, eq. 5).

Reason 2: each word keeps its own information. The residual connection carries the original vector forward, and each sub-layer only adds to it. The attention sub-layer can then concentrate on one job, relating the words to each other, without also having to preserve each word’s embedding and position (StatQuest, “Transformer Neural Networks”, 22:30). Jurafsky and Martin describe the transformer this way, as a **residual stream** (G-1683) per word: it “starts with the original input vector, and the various components read their input from the residual stream and add their output back into the stream” (SLP3 §7.2). If a sub-layer produces a poor output, the word’s earlier vector is still there for the next layer to use.

Without that stream, attention alone tends to make all words alike. Dong et al. (2021) proved that a stack of attention layers without skip connections or feed-forward layers converges to an output in which every word has the same vector, and found that “skip connections play a key role in mitigating” this **rank collapse** (G-1628).

The Notebook measures it on a real IMDB review (its first 30 words, random 64-number embeddings, untrained blocks, mean of 5 random starts). After each block we compute the average cosine similarity between the words (1 means all words point the same way) and between each word and its own input vector (Figure 8).

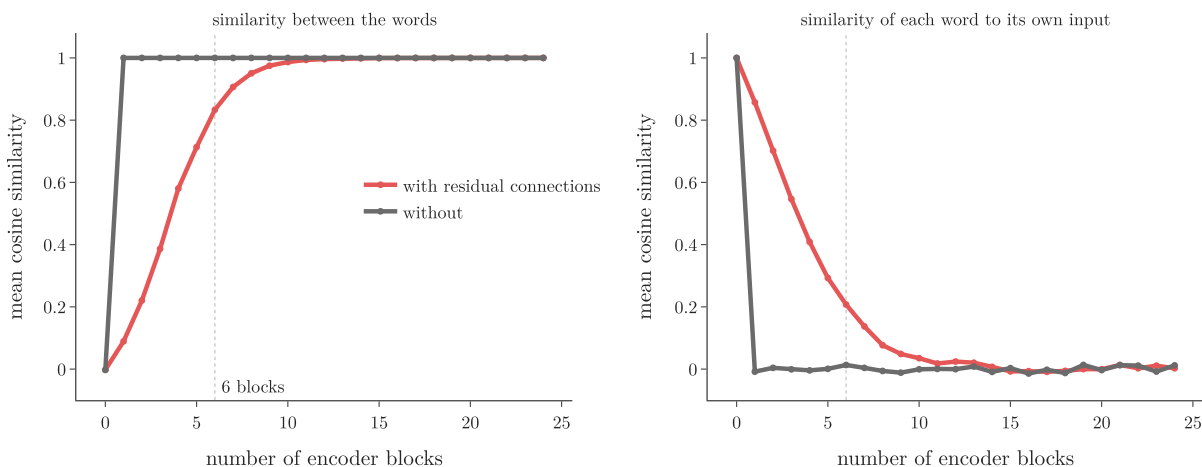


Figure 8: Untrained encoder blocks on the first 30 words of a real review, with and without residual connections, mean of 5 random starts. Left: how alike the words become. Right: how much each word still resembles its own input

Blocks	Between words, without residual	Between words, with residual	Word to own input, without	Word to own input, with
0	0.00	0.00	1.00	1.00
1	1.00	0.09	-0.01	0.86
2	1.00	0.22	0.00	0.70
6	1.00	0.83	0.01	0.21
12	1.00	1.00	0.00	0.02

Without residual connections, one block is enough: all 30 words get the same vector, and that vector has nothing left of any word’s input. With residual connections the words stay distinct and close to their inputs for several blocks. In these untrained blocks the words still drift together by 12 blocks; residual connections slow the collapse rather than stop it, in line with Dong et al.’s “mitigating”. Training then has a signal to work with in every word.

7.2 Why the feed-forward network

Key point: Attention mixes words; the feed-forward network transforms each word on its own, with a ReLU in between. Its ReLU is the block’s only element-wise activation.

Attention already produces contextual vectors, so why add a second sub-layer? Two observations from the formulas:

- **Attention mixes, the feed-forward network transforms.** Each attention output is a weighted average of the value vectors, and each value vector is a linear function of the input ($v = xW_V$). The weights of that average are not linear in the input: they come from a softmax of dot products between the words’ own vectors, so attention as a

whole is not a linear operation. The feed-forward network does not mix positions: it is “position-wise, meaning that it operates on each token position i independently. This makes a contrast with the attention network, whose job is to mix information from different token positions” (SLP3 §7.2.1). Section 5.3 measured this split.

- **A non-linearity per word.** Once the weights are fixed, attention only averages value vectors. The ReLU inside the feed-forward network is the only activation function applied to each word’s own vector in the block (the [activation functions Note](#) explains why stacked layers need one). It lets the block apply a non-linear transformation to each contextual vector.

What the feed-forward layers learn is an open research topic. One finding: in trained language models they behave like **key-value memories**, where the first layer’s weights detect patterns in the input text and the second layer’s weights push the prediction towards particular output words (Geva et al. 2021). Jurafsky and Martin add that the feed-forward parameters “seem to encode most of the factual knowledge in the transformer” (SLP3 §7.2.1).

Extra: Most large language models today replace the ReLU network with a **gated** feed-forward layer, using an activation called SwiGLU, which works better than simpler functions such as ReLU (SLP3 §7.2.1).

7.3 Why 6 blocks

Key point: One block is not enough to represent a language. The paper tried 2, 4, 6 and 8 blocks; 6 scored best, and other models use other numbers.

Language is complex, and a single block has limited power to represent it. Deep learning builds richer representations by stacking layers, each working on the output of the one before; the transformer stacks whole blocks.

The number 6 is an experimental choice. The paper trained versions with different numbers of blocks (Vaswani et al. 2017, Table 3, rows C, English-to-German translation, development set; BLEU measures translation quality):

Blocks N	2	4	6 (base model)	8
BLEU	23.7	25.3	25.8	25.5
Parameters (millions)	36	50	65	80

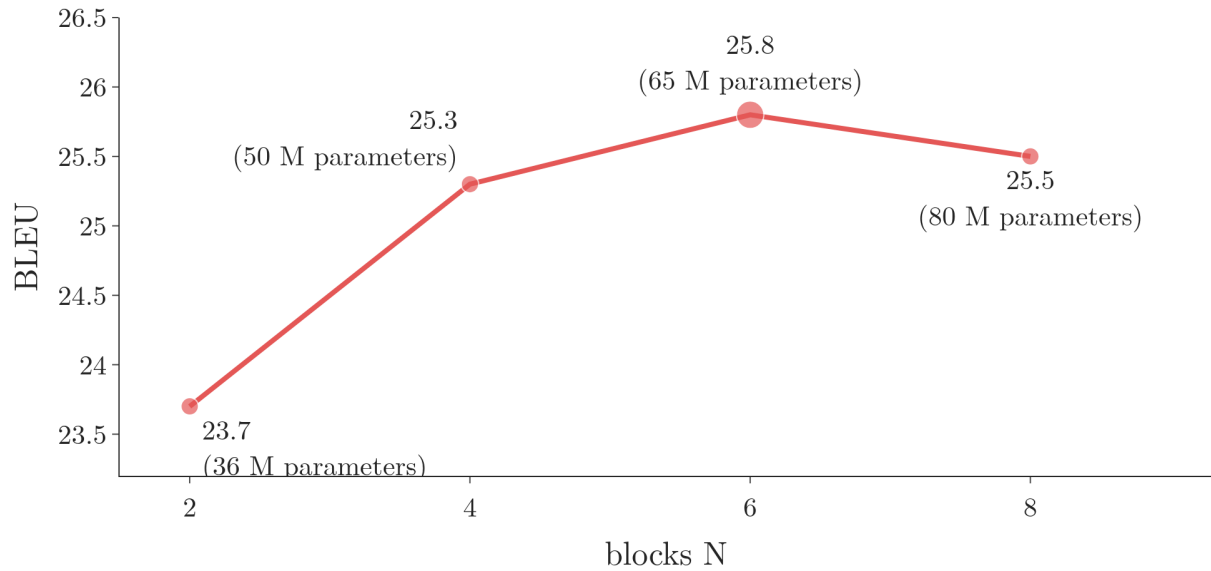


Figure 9: BLEU against the number of blocks N , with each model’s size (Vaswani et al. 2017, Table 3, rows C). The base model, $N = 6$, is the largest dot

Figure 9 draws the table. Going from 2 to 6 blocks adds 2.1 BLEU; going to 8 adds parameters without a gain. The best number depends on the task and the data: BERT-base has 12 blocks and BERT-large 24 (Devlin et al. 2019, §3).

8. Summary

Step	Operation	Shape (3 words)
Input	tokens $\rightarrow$ embeddings + positional encodings	$X: 3 \times 512$
Sub-layer 1	multi-head self-attention	$Z: 3 \times 512$
Add & norm	LayerNorm($X + Z$)	$Z_{\text{norm}}: 3 \times 512$
Sub-layer 2	feed-forward: $\max(0, Z_{\text{norm}}W_1 + b_1)W_2 + b_2$	$H: 3 \times 2048$, then $Y: 3 \times 512$
Add & norm	LayerNorm($Z_{\text{norm}} + Y$)	$Y_{\text{norm}}: 3 \times 512$
Repeat	6 blocks, each with its own weights	3×512 to the decoder

- The encoder is 6 identical blocks; each block is multi-head attention and a position-wise feed-forward network, each followed by add and norm.
- Every vector keeps $d_{\text{model}} = 512$ numbers, which makes the residual additions possible.
- One block has 3,152,384 parameters, two-thirds in the feed-forward network.
- Residual connections give the input and the gradient a direct path through the stack.
- The feed-forward network transforms each word separately, with the block's only ReLU; attention is where words exchange information.

9. Sources

Built from

- CampusX, “Transformer Architecture | Part 1 Encoder Architecture | CampusX”, YouTube, <https://www.youtube.com/watch?v=Vs87qcdm8l0>
- StatQuest with Josh Starmer, “Transformer Neural Networks, ChatGPT’s foundation, Clearly Explained!!!”, YouTube, <https://www.youtube.com/watch?v=zxQyTK8quyY>. 22:30 (residual connections let self-attention relate the words without also having to preserve the embedding and position).
- Sanderson, G. (3Blue1Brown), “Attention in transformers, step-by-step | Deep Learning Chapter 6”, 2024, [3blue1brown.com/lessons/attention](https://www.youtube.com/watch?v=eMlx5fFN), <https://www.youtube.com/watch?v=eMlx5fFN> 14:45–15:44 (the attention output as a change added to the embedding).
- Sanderson, G. (3Blue1Brown), “How might LLMs store facts | Deep Learning Chapter 7”, 2024, [3blue1brown.com/lessons/mlp](https://www.youtube.com/watch?v=9-Jl0dxWQs8), <https://www.youtube.com/watch?v=9-Jl0dxWQs8>. 7:30–13:49 (rows of the first matrix as questions, ReLU as a gate, columns of the second matrix as directions added to the vector).

Other references

- Vaswani, A. et al. (2017). Attention Is All You Need. *NeurIPS 2017*. arXiv:1706.03762. §3.1 (encoder: $N = 6$, two sub-layers, LayerNorm($x + \text{Sublayer}(x)$), $d_{\text{model}} = 512$); §3.3 (position-wise feed-forward network, eq. 2, $d_{\text{ff}} = 2048$); §5.1 (byte-pair encoding); §5.4 (residual dropout, $P_{\text{drop}} = 0.1$); Table 3, rows C.

- Jurafsky, D. and Martin, J. H. *Speech and Language Processing*, 3rd ed. draft (19 August 2026), ch. 7, §7.2 (transformer blocks, residual stream) and §7.2.1 (feedforward layer, SwiGLU).
- He, K., Zhang, X., Ren, S. and Sun, J. (2016). Deep Residual Learning for Image Recognition. *CVPR 2016*. arXiv:1512.03385. §1 (degradation problem); §3.1 (identity mappings); §4.1 (plain vs residual networks).
- He, K., Zhang, X., Ren, S. and Sun, J. (2016). Identity Mappings in Deep Residual Networks. *ECCV 2016*. arXiv:1603.05027. §3, eq. 5 (the gradient’s direct term).
- Dong, Y., Cordonnier, J.-B. and Loukas, A. (2021). Attention is not all you need: pure attention loses rank doubly exponentially with depth. *ICML 2021*. arXiv:2103.03404. Abstract and §1.
- Ba, J. L., Kiros, J. R. and Hinton, G. E. (2016). Layer Normalization. arXiv:1607.06450.
- Geva, M., Schuster, R., Berant, J. and Levy, O. (2021). Transformer Feed-Forward Layers Are Key-Value Memories. *EMNLP 2021*. arXiv:2012.14913. Abstract.
- Devlin, J., Chang, M.-W., Lee, K. and Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL 2019*. arXiv:1810.04805. §3 (BERT-base 12 layers, BERT-large 24).
- Maas, A. L. et al. (2011). Learning Word Vectors for Sentiment Analysis. *ACL 2011*. The IMDB review dataset.

10. Key terms

Term	Meaning
Encoder	The part of the transformer that turns the input sentence into one contextual vector per word
Encoder block	One layer of the encoder: multi-head attention and a feed-forward network, each followed by add and norm; the encoder stacks 6
Sub-layer	One of the two parts of an encoder block: multi-head attention or the feed-forward network
Token	One unit of text the model reads, such as a word or a piece of a word
Tokenisation (G-1983)	Splitting a text into tokens
d_{model}	The number of values in every word vector inside the transformer: 512 in the paper
Residual connection	A path that skips a sub-layer and adds the sub-layer’s input to its output; also called a skip connection
Add and norm	A residual addition followed by layer normalisation: $\text{LayerNorm}(x + \text{Sublayer}(x))$

Term	Meaning
Feed-forward network (FFN)	Two dense layers, $512 \rightarrow 2048$ with ReLU $\rightarrow 512$, inside each block
Position-wise	Applied to each word's vector separately, with the same weights for every position
d_{ff}	The number of hidden nodes of the feed-forward network: 2048 in the paper
Observation	One record of the data; here, one word's vector in a batch
Degradation problem	Deeper plain networks reaching a higher training error than shallower ones
Rank collapse	All word vectors of a sentence becoming the same vector after many attention layers without residual connections